

OCR Further Mathematics
Mechanics - Motion in a Circle
Question Paper
AS and A Level
Further Mathematics A - H235, H245



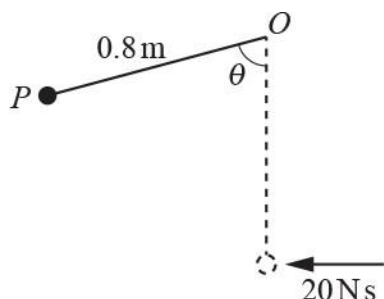
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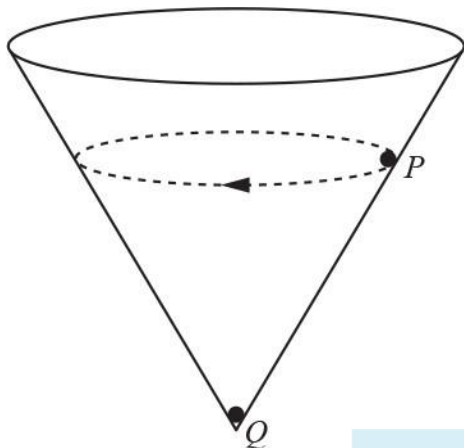
- 1(a) A particle P of mass 4 kg is attached to one end of a light inextensible string of length 0.8 m . The other end of the string is attached to a fixed point O . P is at rest vertically below O when it experiences a horizontal impulse of magnitude 20 N s . In the subsequent motion the angle the string makes with the downwards vertical through O is denoted by θ (see diagram).



Find the magnitude of the acceleration of P at the first instant when $\theta = \frac{1}{3}\pi$ radians. [7]

- (b) Determine the value of θ at which the string first becomes slack. [3]

- 2(a) A hollow cone is fixed with its axis vertical and its vertex downwards. A small sphere P of mass m kg is moving in a horizontal circle on the inner surface of the cone. An identical sphere Q rests in equilibrium inside the cone (see diagram).



The following modelling assumptions are made.

- P and Q are modelled as particles.
- The cone is modelled as smooth.
- There is no air resistance.

Assuming that P moves with a constant speed v m s⁻¹, show that the total mechanical energy of P is $\frac{3}{2}mv^2$ J more than the total mechanical energy of Q . [6]

- (b) Explain how the assumption that P and Q are both particles has been used. [1]

- (c) In practice, P will not move indefinitely in a perfectly circular path, but will actually follow an approximately spiral path on the inside surface of the cone until eventually it collides with Q .

Suggest an improvement that could be made to the model. [1]

- 3(a) One end of a light inextensible string of length 0.75 m is attached to a particle A of mass 2.8 kg. The other end of the string is attached to a fixed point O . A is projected horizontally with speed 6 ms^{-1} from a point 0.75 m vertically above O (see Fig. 3). When OA makes an angle θ with the upward vertical the speed of A is $v\text{ ms}^{-1}$.



Fig. 3

Show that $v^2 = 50.7 - 14.7 \cos \theta$.

[3]

- (b) Given that the string breaks when the tension in it reaches 200 N, find the angle that OA turns through between the instant that A is projected and the instant that the string breaks. [4]

4(a)



The flat surface of a smooth solid hemisphere of radius r is fixed to a horizontal plane on a planet where the acceleration due to gravity is denoted by g . O is the centre of the flat surface of the hemisphere.

A particle P is held at a point on the surface of the hemisphere such that the angle between OP and the upward vertical through O is α , where $\cos \alpha = \frac{3}{4}$.

P is then released from rest. F is the point on the plane where P first hits the plane (see diagram).

Find an exact expression for the distance OF . [11]

- (b) The acceleration due to gravity on and near the surface of the planet Earth is roughly 6y .

Explain whether OF would increase, decrease or remain unchanged if the action were repeated on the planet Earth.

[1]



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- 5(a) Two particles A and B , of masses m kg and 1 kg respectively, are connected by a light inextensible string of length d m and placed at rest on a smooth horizontal plane a distance of $\frac{1}{2}d$ m apart. B is then projected horizontally with speed v m s⁻¹ in a direction perpendicular to AB .

Show that, at the instant that the string becomes taut, the magnitude of the instantaneous impulse in the

string, I N s, is given by $I = \frac{\sqrt{3}mv}{2(1+m)}$.



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[4]

- (b) Find, in terms of m and v , the kinetic energy of B at the instant after the string becomes taut. Give your answer as a single algebraic fraction.

[3]



- (c) In the case where m is very large, describe, with justification, the approximate motion of B after the string becomes taut.

[2]

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- 6(a) A right circular cone C of height 4 m and base radius 3 m has its base fixed to a horizontal plane. One end of a light elastic string of natural length 2 m and modulus of elasticity 32 N is fixed to the vertex of C . The other end of the string is attached to a particle P of mass 2.5 kg.

P moves in a horizontal circle with constant speed and in contact with the smooth curved surface of C . The extension of the string is 1.5 m.

Find the tension in the string.

[2]



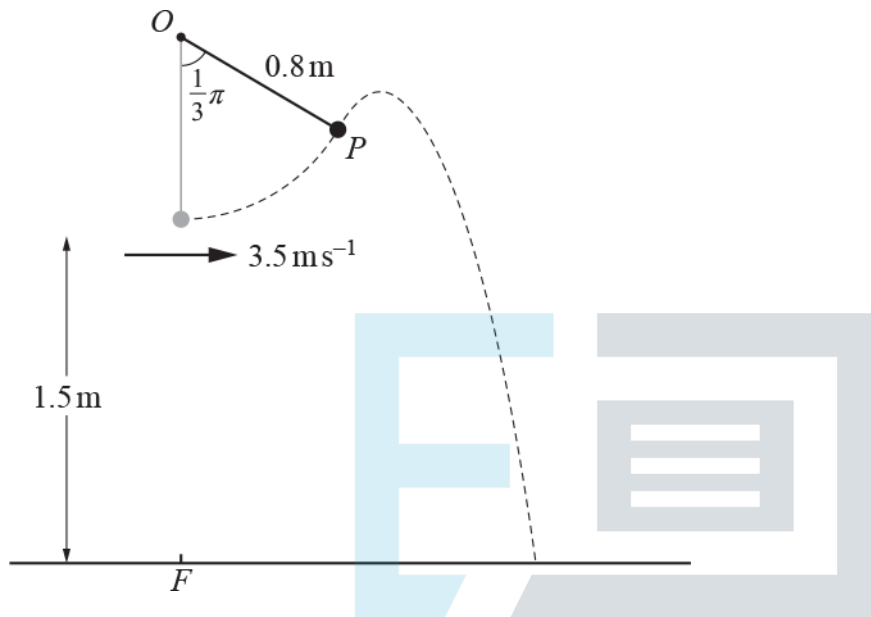
(b) Find the speed of P . [7]

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- 7 One end of a light inextensible string of length 0.8 m is attached to a fixed point, O . The other end is attached to a particle P of mass 1.2 kg. P hangs in equilibrium at a distance of 1.5 m above a horizontal plane. The point on the plane directly below O is F .

P is projected horizontally with speed 3.5 ms^{-1} . The string breaks when OP makes an angle of $\frac{1}{3}\pi$ radians with the downwards vertical through O (see diagram).



- (a) Find the magnitude of the tension in the string at the instant before the string breaks.

[4]

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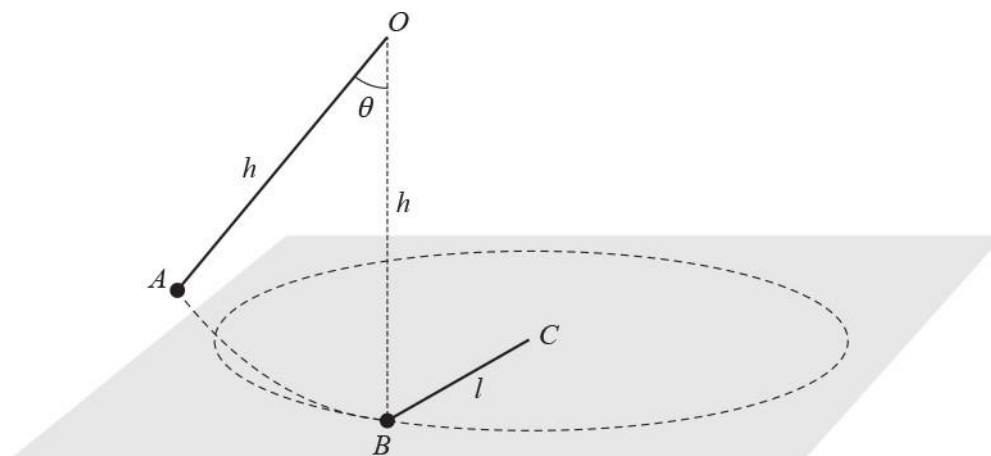
(b) Find the distance between F and the point where P first hits the plane.

[7]



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- 8 A point O is situated a distance h above a smooth horizontal plane, and a particle A of mass m is attached to O by a light inextensible string of length h . A particle B of mass $2m$ is at rest on the plane, directly below O , and is attached to a point C on the plane, where $BC = l$, by a light inextensible string of length l . A is released from rest with the string OA taut and making an acute angle θ with the downward vertical (see diagram).



A moves in a vertical plane perpendicular to CB and collides directly with B . As a result of this collision, A is brought to rest and B moves on the plane in a horizontal circle with centre C . After B has made one complete revolution the particles collide again.

- (a) Show that, on the next occasion that A comes to rest, the string OA makes an angle ϕ with the

downward vertical through O , where $\cos \phi = \frac{3 + \cos \theta}{4}$.

[9]

A and B collide again when AO is next vertical.

(b) Find the percentage of the original energy of the system that remains immediately after this collision.

[5]



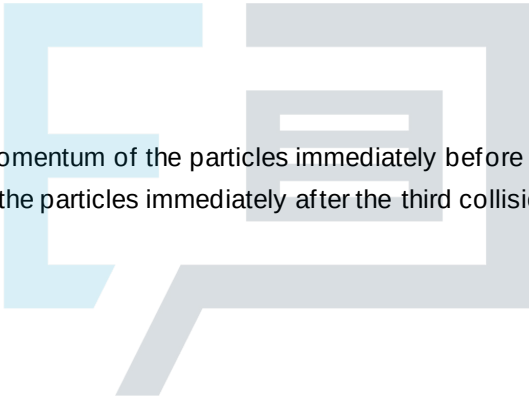
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- (c) Explain why the total momentum of the particles immediately before the first collision is the same as the total momentum of the particles immediately after the second collision. [1]

- (d) Explain why the total momentum of the particles immediately before the first collision is different from the total momentum of the particles immediately after the third collision. [1]



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9 Assume that the earth moves round the sun in a circle of radius 1.50×10^8 km at constant speed, with one complete orbit taking 365 days. Given that the mass of the earth is 5.97×10^{24} kg,

(a) calculate the magnitude of the force exerted by the sun on the earth, giving your answer in newtons,

[5]



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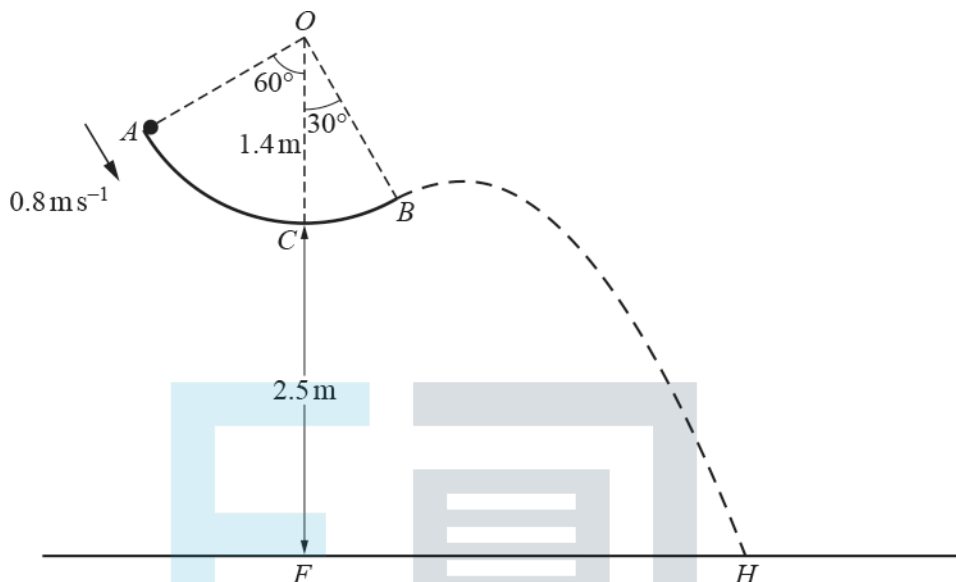
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(b) state the direction in which this force acts.

[1]

- 10 A smooth track AB is in the shape of an arc of a circle with centre O and radius 1.4 m . The track is fixed in a vertical plane with A above the level of B and a point C on the track vertically below O . Angle AOC is 60° and angle COB is 30° . Point C is 2.5 m vertically above the point F , which lies in a horizontal plane. A particle of mass 0.4 kg is placed at A and projected down the track with an initial velocity of 0.8 m s^{-1} . The particle first hits the plane at point H (see diagram).



- (a) Find the magnitude of the contact force between the particle and the track when the particle is at B .

[5]

(b) Find the distance FH .

[7]

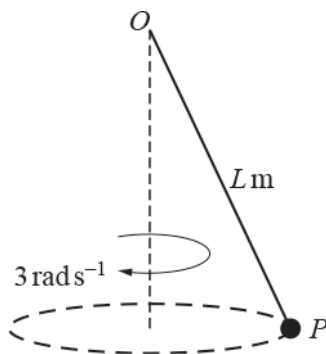


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- 11 A particle P of mass 3.5 kg is attached to one end of a light elastic string of natural length 0.8 m and modulus of elasticity 75 N . The other end of the string is attached to a fixed point O . The particle rotates in a horizontal circle with a constant angular velocity of 3 rad s^{-1} . The centre of the circle is vertically below O . The magnitude of the tension in the string is $T\text{ N}$ and the length of the extended string is $L\text{ m}$ (see diagram).



- (a) By considering the acceleration of P , show that $T = 31.5 L$.

[4]

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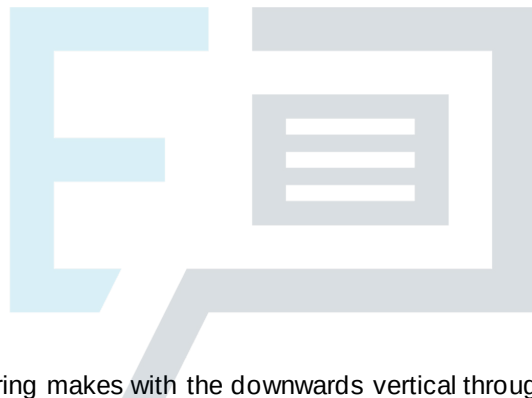
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- (b) Write down another relationship between T and L .

[1]

(c) Find the value of T and the value of L .

[3]



(d) Find the angle that the string makes with the downwards vertical through O .

[2]

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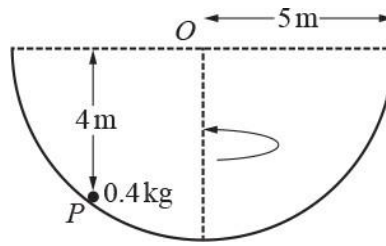


Fig. 1

A particle P of mass 0.4 kg is moving on the smooth inner surface of a fixed hollow hemisphere which has centre O and radius 5 m . P moves with constant angular speed in a horizontal circle at a vertical distance of 4 m below the level of O (see Fig. 1).

- (i) Find, in terms of g , the magnitude of the force exerted by the hemisphere on P .

[3]

- (ii) Show that the time taken, in seconds, for P to complete one revolution of its circular path is given by

$$\frac{4\pi}{\sqrt{g}}.$$

[4]

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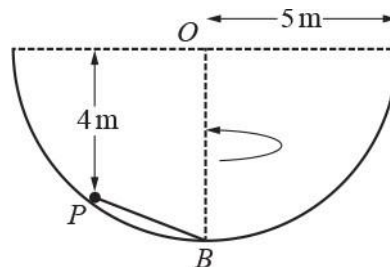


Fig. 2

One end of a light string is now attached to P . The other end of the string is attached to the lowest point B of the hemisphere. P moves in the same horizontal circle as before (see Fig. 2).

- (iii) Given that the tension in the string is $\sqrt{40}$ N, calculate the new speed of P . [6]



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- 13 A uniform rod with centre C has mass $2M$ and length $4a$. The rod is free to rotate in a vertical plane about a smooth fixed horizontal axis passing through a point A on the rod, where $AC = ka$ and $0 < k < 2$. The rod is making small oscillations about the equilibrium position with period T .

- (i) Show that $T = 2\pi\sqrt{\frac{a}{3g}\left(\frac{4+3k^2}{k}\right)}$. (You may assume the standard formula $T = 2\pi\sqrt{\frac{I}{mgh}}$ for the period of small oscillations of a compound pendulum.) [4]

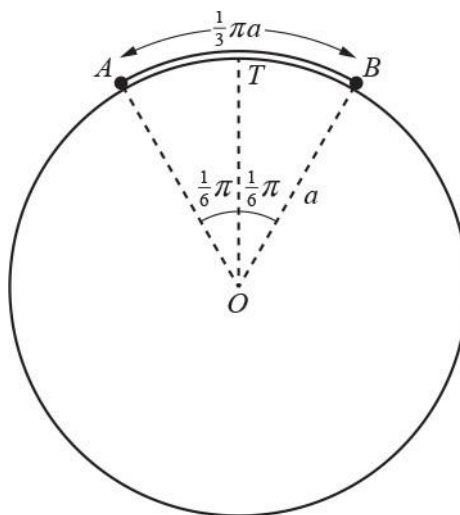


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- (ii) Hence find the value of k^2 for which the period of oscillations is least. [3]



Particle A of mass $3m$ and particle B of mass $2m$ are joined by a light inextensible string of length $\frac{1}{3}\pi a$. The particles and the string rest in a vertical plane on the surface of a smooth cylinder, of radius a , which has its axis horizontal; O is the centre of the vertical cross-section of the cylinder containing the particles and T is the uppermost point on the surface of the cylinder. The string is taut and radii OA and OB each make an angle of $\frac{1}{6}\pi$ radians with OT (see diagram). A and B are released from rest, and after t seconds angle AOT is $(\frac{1}{6}\pi + \theta)$ radians and both A and B are moving with speed v .

- (i) Show that, while A and B remain in contact with the cylinder,

$$v^2 = \frac{2}{5}ag \left(5 \cos \frac{1}{6}\pi - 3 \cos \left(\frac{1}{6}\pi + \theta \right) - 2 \cos \left(\frac{1}{6}\pi - \theta \right) \right). \quad [4]$$

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- (ii) Find an expression in terms of m and g for the force exerted on A by the cylinder at the instant when B is passing through T . [4]

- 15 A light inextensible taut rope, of length 4 m, is attached at one end A to the centre of the horizontal ceiling of a gym. The other end of the rope B is being held by a child of mass 35 kg. Initially the child is held at rest with the rope making an angle of 60° to the downward vertical and it may be assumed that the child can be modelled as a particle attached to the end of the rope. The child is released at a height 5 m above the horizontal ground.

(a) Show that the speed, $v \text{ m s}^{-1}$, of the child when the rope makes an angle θ with the downward vertical

is given by $v^2 = 4g(2\cos\theta - 1)$.

[5]



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- (b) At the instant when $\theta = 0^\circ$, the child lets go of the rope and moves freely under the influence of gravity only. Determine the speed and direction of the child at the moment that the child reaches the ground. [5]

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- (c) The child returns to the initial position and is released again from rest. Find the value of θ when the tension in the rope is three times greater than the tension in the rope at the instant the child is released.

[5]



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- 16 A uniform square lamina, of mass 5 kg and side 0.2 m, is rotating about a fixed vertical axis that is perpendicular to the lamina and that passes through its centre. A couple of constant moment 0.06 N m is applied to the lamina. The lamina turns through an angle of 155 radians while its angular speed increases from 8 rad s^{-1} to $\omega \text{ rad s}^{-1}$. Find ω .

[4]

17 A turntable is rotating at 3 rad s^{-1} . The turntable is then accelerated so that after 4 revolutions it is rotating at 12.4 rad s^{-1} . Assuming that the angular acceleration of the turntable is constant,

(i) find the angular acceleration, [3]

(ii) find the time taken to increase its angular speed from 3 rad s^{-1} to 12.4 rad s^{-1} . [2]



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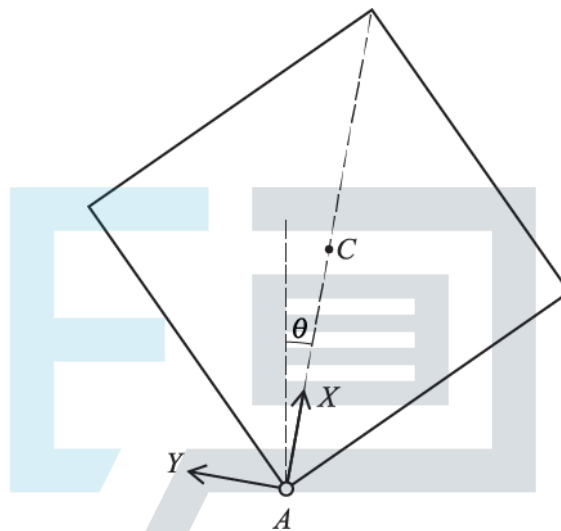
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18 A uniform square lamina has mass m and sides of length $2a$.

- (i) Calculate the moment of inertia of the lamina about an axis through one of its corners perpendicular to its plane.

[3]



The uniform square lamina has centre C and is free to rotate in a vertical plane about a fixed horizontal axis passing through one of its corners A . The lamina is initially held such that AC is vertical with C above A . The lamina is slightly disturbed from rest from this initial position. When AC makes an angle θ with the upward vertical, the force exerted by the axis on the lamina has components X parallel to AC and Y perpendicular to AC (see diagram).

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- (ii) Show that the angular speed, ω , of the lamina satisfies $a\omega^2 = \frac{3}{4}g\sqrt{2}(1 - \cos\theta)$.

[4]

(iii) Find X and Y in terms of m , g and θ .

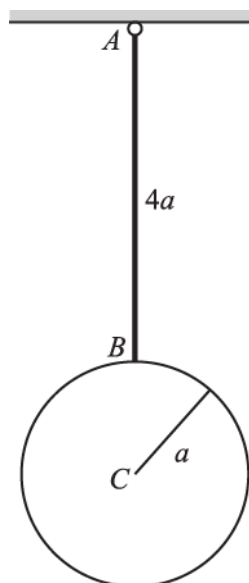
[6]



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A pendulum consists of a uniform rod AB of length $4a$ and mass $4m$ and a spherical shell of radius a , mass m and centre C . The end B of the rod is rigidly attached to a point on the surface of the shell in such a way that ABC is a straight line. The pendulum is initially at rest with B vertically below A and it is free to rotate in a vertical plane about a smooth fixed horizontal axis passing through A (see diagram).

- (i) Show that the moment of inertia of the pendulum about the axis of rotation is $47ma^2$.

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[4]

A particle of mass m is moving horizontally in the plane in which the pendulum is free to rotate. The particle has speed $\sqrt{kga}$, where k is a positive constant, and strikes the rod at a distance $3a$ from A . In the subsequent motion the particle adheres to the rod and the combined rigid body P starts to rotate.

- (ii) Show that the initial angular speed of P is $\frac{3}{56}\sqrt{\frac{kg}{a}}$.

[4]

(iii) For the case $k = 4$, find the angle that P has turned through when P first comes to instantaneous rest.

[4]



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(iv) Find the least value of k such that the rod reaches the horizontal.



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[2]

- 20 A camshaft inside an engine is rotating with angular speed 42 rad s^{-1} . When the throttle is opened the camshaft speeds up with constant angular acceleration, and 8 seconds after the throttle was opened the angular speed is 76 rad s^{-1} .

(i) Find the angular acceleration of the camshaft.



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[2]

(ii) Find the time taken for the camshaft to turn through 810 radians from the moment that the throttle was opened.

[3]

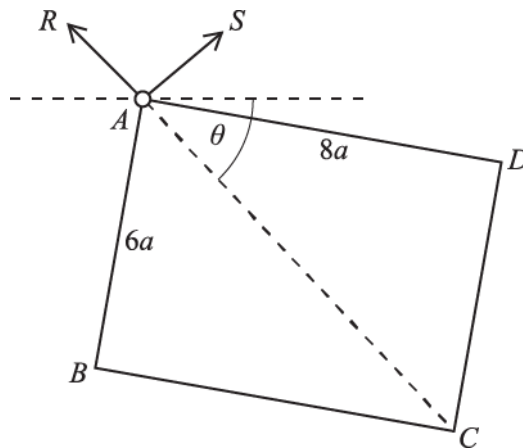


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21



$ABCD$ is a uniform rectangular lamina with mass m and sides $AB = 6a$ and $AD = 8a$. The lamina rotates freely in a vertical plane about a fixed horizontal axis passing through A , and it is released from rest in the position with D vertically above A . When the diagonal AC makes an angle θ below the horizontal, the force acting on the lamina at A has components R parallel to CA and S perpendicular to CA (see diagram).

- (i) Find the moment of inertia of the lamina about the axis through A , in terms of m and a .

[3]

- (ii) Show that the angular speed of the lamina is $\sqrt{\frac{3g(4 + 5 \sin \theta)}{50a}}$.

[3]

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(iii) Find the angular acceleration of the lamina, in terms of a , g and θ .



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[2]

(iv) Find R and S , in terms of m , g and θ .

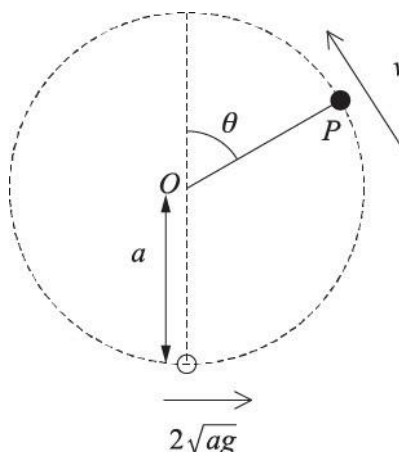
[6]



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One end of a light inextensible string of length a is attached to a fixed point O . A particle P of mass m is attached to the other end of the string and hangs at rest. P is then projected horizontally from this position with speed $2\sqrt{ag}$. When the string makes an angle θ with the upward vertical P has speed v (see diagram). The tension in the string is T .

- (i) Find an expression for T in terms of m , g and θ , and hence find the height of P above its initial level when the string becomes slack.

[6]

P is now projected horizontally from the same initial position with speed U .

- (ii) Find the set of values of U for which the string does not remain taut in the subsequent motion.

[5]

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- 23 A smooth cylinder of radius a m is fixed with its axis horizontal and O is the centre of a cross-section. Particle P , of mass 0.4 kg, and particle Q , of mass 0.6 kg, are connected by a light inextensible string of length πa m. The string is held at rest with P and Q at opposite ends of the horizontal diameter of the cross-section through O (see Fig. 1). The string is released and Q begins to descend. When OP has rotated through θ radians, with P remaining in contact with the cylinder, the speed of each particle is v m s⁻¹ (see Fig. 2).

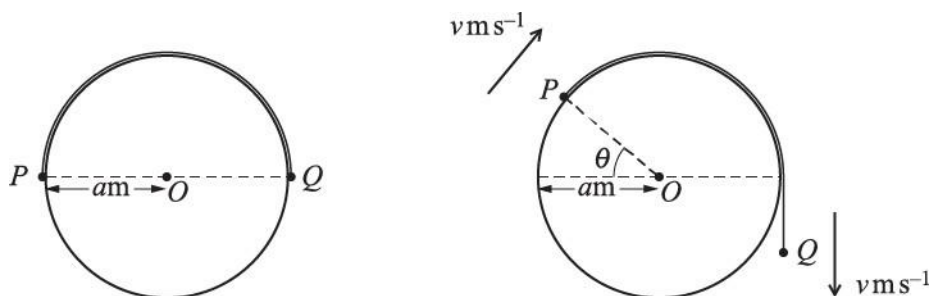


Fig. 1

Fig. 2

- (i) Show that $v^2 = 3.92a(3\theta - 2\sin \theta)$ and find an expression in terms of θ for the normal force of the cylinder on P at this time.

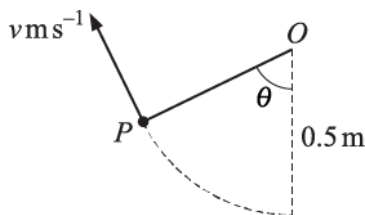
[9]

- (ii) Given that P leaves the surface of the cylinder when $\theta = \alpha$, show that $\sin \alpha = k\alpha$ where k is a constant to be found.

[2]

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One end of a light inextensible string of length 0.5 m is attached to a fixed point O . A particle P of mass 0.2 kg is attached to the other end of the string. P is projected horizontally from the point 0.5 m below O with speed $u \text{ ms}^{-1}$. When the string makes an angle of θ with the downward vertical the particle has speed $v \text{ ms}^{-1}$ (see diagram).

- (i) Show that, while the string is taut, the tension, $T \text{ N}$, in the string is given by

$$T = 5.88 \cos \theta + 0.4u^2 - 3.92.$$

[5]

- (ii) Find the least value of u for which the particle will move in a complete circle.

- (iii) If in fact $u = 3.5 \text{ ms}^{-1}$, find the speed of the particle at the point where the string first becomes slack.

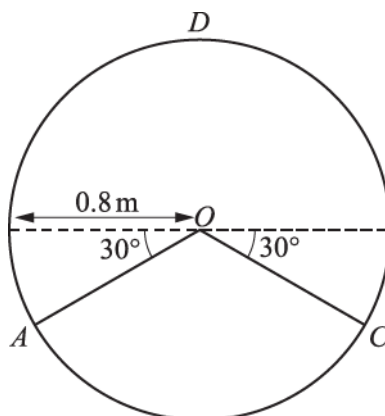
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[4]



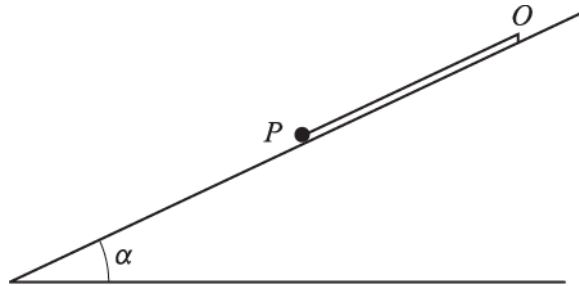
A hollow cylinder is fixed with its axis horizontal. O is the centre of a vertical cross-section of the cylinder and D is the highest point on the cross-section. A and C are points on the circumference of the cross-section such that AO and CO are both inclined at an angle of 30° below the horizontal diameter through O . The inner surface of the cylinder is smooth and has radius 0.8 m (see diagram). A particle P , of mass $m\text{ kg}$, and a particle Q , of mass $5m\text{ kg}$, are simultaneously released from rest from A and C , respectively, inside the cylinder. P and Q collide; the coefficient of restitution between them is 0.95 .

- (i) Show that, immediately after the collision, P moves with speed 6.3 m s^{-1} , and find the speed and direction of motion of Q .

Exam Papers Practice [8]

- (ii) Find, in terms of m , an expression for the normal reaction acting on P when it subsequently passes through D .

[6]



One end of a light elastic string, of natural length 0.3 m , is attached to a fixed point O on a smooth plane that is inclined at an angle α to the horizontal, where $\sin \alpha = 0.2$. A particle P of mass $m\text{ kg}$ is attached to the other end of the string. The string lies along a line of greatest slope of the plane and has modulus of elasticity $2.45m\text{ N}$ (see diagram).

- (i) Show that in the equilibrium position the extension of the string is 0.24 m .

[2]

P is given a velocity of 0.3 m s^{-1} down the plane from the equilibrium position.

- (ii) Show that P performs simple harmonic motion with period 2.20 s (correct to 3 significant figures), and find the amplitude of the motion.

[6]



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(iii) Find the distance of P from O and the velocity of P at the instant 1.5 seconds after P is set in motion.

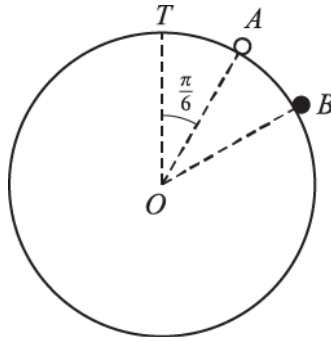
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A fixed smooth sphere of radius 0.6 m has centre O and highest point T . A particle of mass m kg is released from rest at a point A on the sphere, such that angle TOA is $\frac{\pi}{6}$ radians. The particle leaves the surface of the sphere at B (see diagram).

(i) Show that $\cos TOB = \frac{\sqrt{3}}{3}$

[6]



(ii) Find the speed of the particle at B .

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[2]

(iii) Find the transverse acceleration of the particle at B .

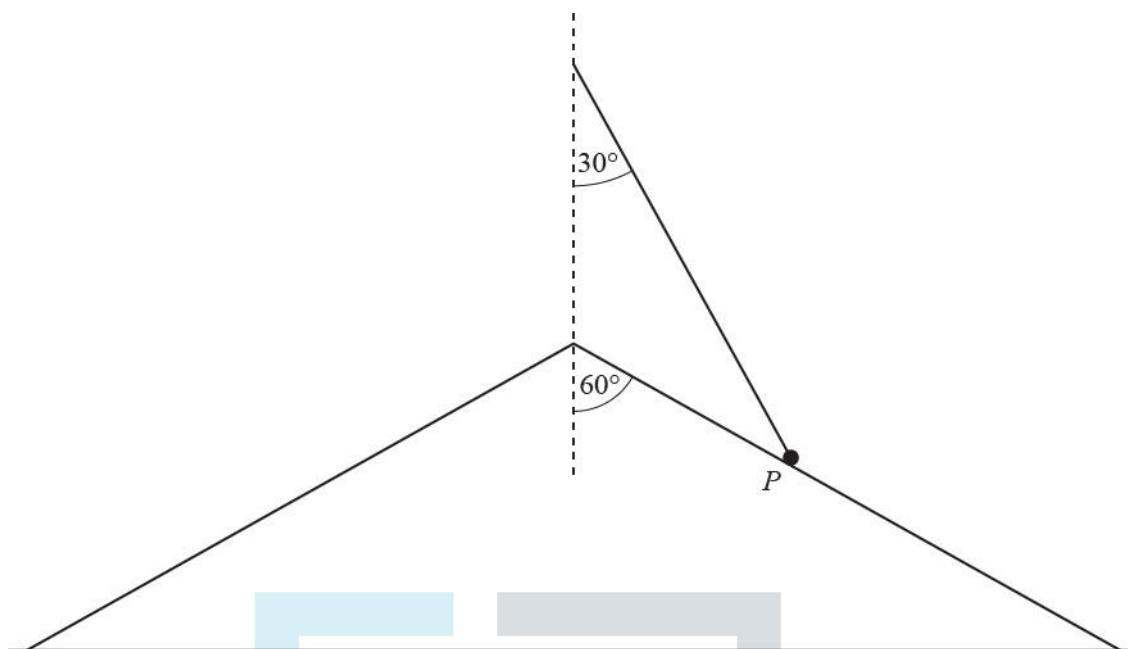
[2]



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A smooth solid cone of semi-vertical angle 60° is fixed to the ground with its axis vertical. A particle P of mass m is attached to one end of a light inextensible string of length a . The other end of the string is attached to a fixed point vertically above the vertex of the cone. P rotates in a horizontal circle on the surface of the cone with constant angular velocity ω . The string is inclined to the downward vertical at an angle of 30° (see diagram).

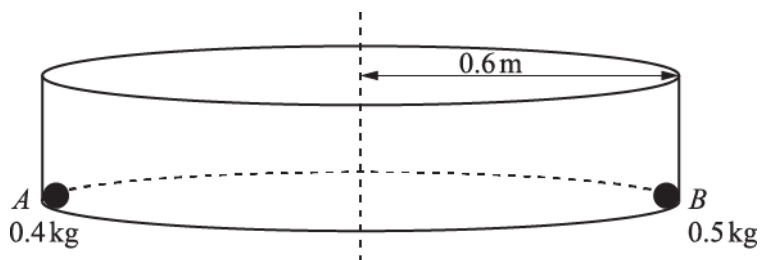
- (i) Show that the magnitude of the contact force between the cone and the particle is [6]
- (ii) Given that $a = 0.5$ m and $m = 3.5$ kg, find, in either order, the greatest speed for which the particle remains in contact with the cone and the corresponding tension in the string. [3]

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- 29 A particle of mass 0.3 kg is attached to one end of a light inextensible string. The other end of the string is attached to a fixed point A . The particle moves in a horizontal circle of radius 0.343 m, with centre vertically below A , at a constant angular speed of 6 rad s^{-1} . Find the tension in the string and the angle at which the string is inclined to the vertical.

[6]



Two small spheres, A and B , are free to move on the inside of a smooth hollow cylinder, in such a way that they remain in contact with both the curved surface of the cylinder and its horizontal base. The mass of A is 0.4 kg , the mass of B is 0.5 kg and the radius of the cylinder is 0.6 m (see diagram). The coefficient of restitution between A and B is 0.35 . Initially, A and B are at opposite ends of a diameter of the base of the cylinder with A travelling at a constant speed of $v\text{ ms}^{-1}$ and B stationary. The magnitude of the force exerted on A by the curved surface of the cylinder is 6 N .

- (i) Show that $v = 3$.

[2]

(ii) Calculate the speeds of the particles after A 's first impact with B .

[6]



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Sphere B is removed from the cylinder and sphere A is now set in motion with constant angular speed $\omega \text{ rad s}^{-1}$. The magnitude of the total force exerted on A by the cylinder is 4.9 N .

(iii) Find ω .

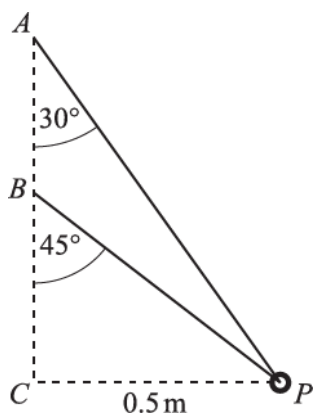
[4]



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A small smooth ring P of mass 0.4 kg is threaded onto a light inextensible string fixed at A and B as shown in the diagram, with A vertically above B . The string is inclined to the vertical at angles of 30° and 45° at A and B respectively. P moves in a horizontal circle of radius 0.5 m about a point C vertically below B .

- (i) Calculate the tension in the string.

[3]

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- (ii) Calculate the speed of P .

[3]

The end of the string at B is moved so both ends of the string are now fixed at A .

(iii) Show that, when the string is taut, AP is now 0.854 m correct to 3 significant figures.

[2]



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P moves in a horizontal circle with angular speed 3.46 rad s^{-1} .

(iv) Find the tension in the string and the angle that the string now makes with the vertical.

[4]



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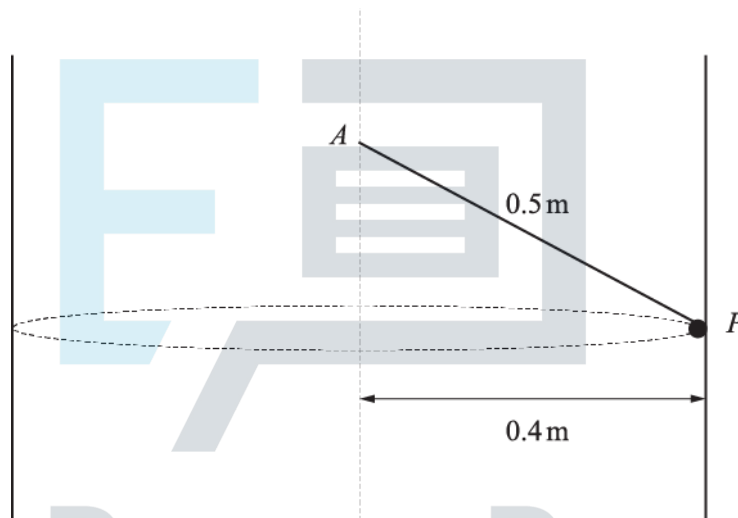
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- 32 A vertical hollow cylinder of radius 0.4 m is rotating about its axis. A particle P is in contact with the rough inner surface of the cylinder. The cylinder and P rotate with the same constant angular speed. The coefficient of friction between P and the cylinder is μ .

- (i) Given that the angular speed of the cylinder is 7 rad s^{-1} and P is on the point of moving downwards, find the value of μ .

[5]

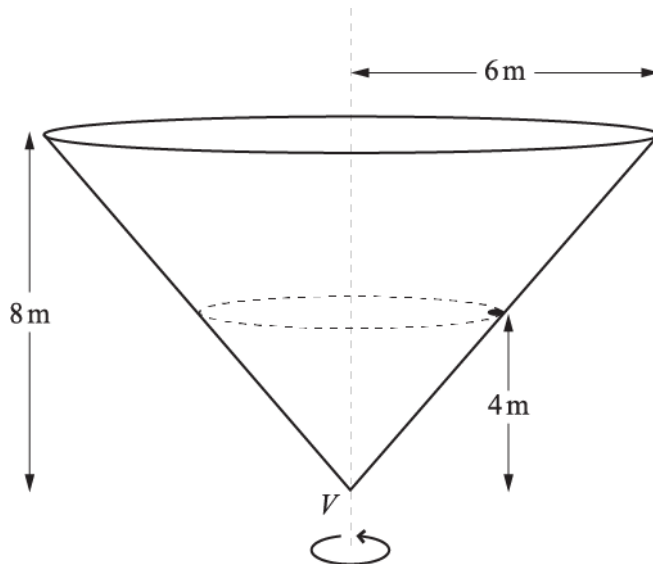
The particle is now attached to one end of a light inextensible string of length 0.5 m . The other end is fixed to a point A on the axis of the cylinder (see diagram).



- (ii) Find the angular speed for which the contact force between P and the cylinder becomes zero.

[5]

33



A conical shell has radius 6 m and height 8 m. The shell, with its vertex V downwards, is rotating about its vertical axis. A particle, of mass 0.4 kg, is in contact with the rough inner surface of the shell. The particle is 4 m above the level of V (see diagram). The particle and shell rotate with the same constant angular speed. The coefficient of friction between the particle and the shell is μ .

- (i) The frictional force on the particle is FN , and the normal force of the shell on the particle is RN . It is given that the speed of the particle is 4.5 ms^{-1} , which is the smallest possible speed for the particle not to slip.

(a) By resolving vertically, show that $4F + 3R = 19.6$.

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[2]

- (b) By finding another equation connecting F and R , find the values of F and R and show that $\mu = 0.336$, correct to 3 significant figures.

[6]



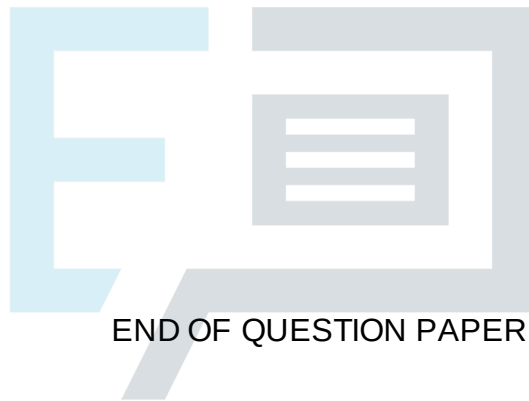
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(ii) Find the largest possible angular speed of the shell for which the particle does not slip.

[6]



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